YUNUS EMRE ULUÇ

SOFTWARE ENGINEER · AI & PYTHON BACKEND

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ABOUT ME

I am a **Software Engineer** who graduated **top of my faculty and department**, and I work across the whole stack an AI product actually needs — PyTorch training, FastAPI/WebSocket backends, Next.js frontends, cross-platform mobile. I do not stop at the prototype: the systems I build run, are tested and are used. AI tooling and autonomous agent workflows sit **at the centre of my daily output** — I run parallel multi-agent lanes, then verify and integrate every result myself.

I build **AI systems end to end**. On **Chiron** I designed a 36-tool LLM agent architecture with its own MCP server (42 pytest tests), on-device computer vision, a two-step approval layer for irreversible actions and desktop + Android clients — shipped solo. On **FractureAI**, a TÜBİTAK-funded medical diagnostics project, I owned the entire pipeline: ~67,000 X-rays, a three-model CNN + transformer ensemble at **91.54% accuracy** (Cohen's κ 0.831, AUC 0.960), Grad-CAM explainability and the clinician-facing web application.

EXPERIENCE

Software Engineering Intern — Opak Yazılım, Konya — Jul 2025

Mobile & Cross-Platform Development · engineering internship · application delivered and presented

- Built and delivered **NICEO & NEO**, cross-platform mobile e-commerce applications (Flutter/Dart and React Native/TypeScript) covering product catalogue, cart and checkout flows.
- Integrated REST APIs behind a layered, UML-modelled architecture with explicit state management; presented the finished application to the company at the end of the internship.

AI & AGENT ENGINEERING

- **Training & evaluation discipline:** patient-level splits against leakage, Focal Loss for class imbalance, Confident Learning for label noise, stratified K-fold and 10× TTA — the difference between a notebook number and a number that survives review.
- **Multi-agent workflows:** I split independent work into parallel Claude Code and Codex CLI lanes, isolate them with git worktrees, then verify and integrate the output myself; recurring pipelines are packaged as custom agent skills (~6,000 lines). Judgment-heavy work stays on top-tier models, mechanical execution is delegated to cheaper tiers.

SELECTED PROJECTS & SYSTEMS BUILT

Chiron — LLM Agent Platform & MCP Server · Desktop & Android — 2026

Sole Developer · Python + Next.js + Kotlin · github.com/yunusemreuluc/Chiron

- Designed a **36-tool agent architecture** on the Gemini Live API, covering tool schemas, routing, fault tolerance and multi-step task execution.
- Added an **MCP (Model Context Protocol) server** with 4 tools on the official Python SDK, covered by **42 pytest tests** through a real MCP client and connected to Claude Code; a reliability and safety review led to 7 of 8 findings fixed (input validation, personal-data masking, resource limits).
- Ran object recognition (YOLO-World) and OCR (EasyOCR, TR+EN) **fully on-device**, removing API quota dependency and network latency.
- Built a **two-step confirmation layer** for screen analysis and mouse/keyboard control — no irreversible action executes without explicit user approval, and deletions route to the recycle bin.
- Connected a FastAPI + WebSocket backend to a React Three Fiber 3D interface streaming agent state in real time (Next.js 16, React 19, TypeScript, Zustand); added a persistent memory layer and proactive background triggers.
- Wrote the **Android client** in Kotlin + Jetpack Compose with a custom-trained wake-word model running **fully offline on ONNX Runtime**; the PC link is secured with a Tailscale private network and PIN.

Dilcep — Offline-First Translation & Language-Learning App (Android) 2026

Sole Developer · Expo / React Native, TypeScript, Kotlin

- Translates text, speech and camera images **on the device** (works in airplane mode) through **three native Kotlin modules** (ML Kit translation, speech recognition, OCR).
- **191 English lessons (A1–B1)** and a 4,368-word FSRS vocabulary path; SQLite + Supabase sync.

FractureAI — TÜBİTAK 2209-A Research Project 2025 – 2026

Project Lead · deep learning based fracture detection in X-ray imaging · TÜBİTAK is Türkiye's national scientific research council · github.com/yunusemreuluc/xray-fracture-detection

- Built a data pipeline over **~67,000 X-ray images** merged from 4 open datasets (MURA, GRAZPEDWRI-DX, FracAtlas, Multi-Region), using patient-level stratified splitting to eliminate leakage.
- Designed a kappa-weighted soft-voting ensemble of EfficientNet-B5, ConvNeXt-Base and Swin Transformer V2-Base, reaching **91.54% accuracy, Cohen's Kappa 0.831 and AUC 0.960** with 10× TTA.
- Normalized clinical image variance (CLAHE, polarity normalization, automatic annotation masking) and shipped the full web application (FastAPI + Three.js) giving clinicians real-time predictions with **Grad-CAM** heatmaps; authored the grant proposal, final report and thesis.

Dengue Fever Detection — Hematological Data Analysis 2026

Researcher · machine learning based diagnostic study — under review for publication · github.com/yunusemreuluc/dengue-fever-detection

- Benchmarked **9 algorithms** (Random Forest, XGBoost, Gradient Boosting, SVM, MLP and others) across 1,523 patient records with 19 hematological parameters, under Stratified K-Fold cross-validation with three feature-selection strategies.

TECHNICAL SKILLS

Languages	Python · TypeScript · JavaScript · Kotlin · C++ · C# · Dart · SQL · MATLAB
AI / ML	PyTorch · torchvision · scikit-learn · XGBoost · Albumentations · Pandas · NumPy · Ultralytics (YOLO-World) · EasyOCR · ONNX Runtime · ML Kit
LLM & Agents	Gemini Live API · function calling · MCP (Model Context Protocol) · tool schema design · prompt engineering · multi-agent orchestration · Claude Code · Codex CLI
Computer Vision	EfficientNet · ConvNeXt · Swin Transformer · transfer learning · ensembling & soft-voting · TTA · Grad-CAM · CLAHE · Focal Loss · OpenCV
Backend & Web	FastAPI · Django · Uvicorn · REST APIs · WebSockets · PostgreSQL · MySQL · Next.js · React · Tailwind CSS · Three.js / React Three Fiber · WordPress / WooCommerce · SQLite · Supabase
Mobile	Kotlin · Jetpack Compose · React Native / Expo · Flutter (Dart) · state management · REST integration
Embedded	Arduino · ESP32 · MPU6050 · nRF24L01 · MQTT · robot kinematics (DH parameters)
Tools	Git & GitHub · pytest · unit & integration testing · Google Colab (A100) · Jupyter · VS Code · Jira · UML · ReportLab · Pillow

EDUCATION

B.Sc. Software Engineering — Muş Alparslan University 2022 – 2026

- **Valedictorian of the faculty and the department** — graduated with the highest GPA.
- Thesis: "A Deep Learning Based AI System for Fracture and Dislocation Detection in Clinical X-Ray Images".

CERTIFICATIONS & LANGUAGES

CCNA: Introduction to Networks — Cisco Networking Academy Jan 2025

Turkish — Native · **English** — B1 (CEFR) · reads technical documentation and academic literature

ADDITIONAL PROJECTS

Digital product pipeline (Python — image-to-PDF delivery in one command, 39 product sets) · **Gesture-controlled smart car** (C++, Arduino, nRF24L01) · **Türkiye Museum Guide** (Django) — github.com/yunusemreuluc